

sensor (DHT11), 5V DC soil moisture sensor, ultrasonic sensor (HC-SR04) and a few other components.

In this project, Load3 (Light1) and Load2 (Light2) are controlled through RL3 and RL2, respectively. DHT11 sensor is used to monitor room temperature and humidity. In Blynk app, virtual levels V5 and V6 are used to display humidity and temperature, respectively. For that, we used `Blynk.virtualWrite(V5, h)` and `Blynk.virtualWrite(V6, t)` in the code.

Soil moisture sensor is used to detect garden soil moisture level. If soil is dry, the garden motor pump turns on automatically to pump out water for the plant.

Ultrasonic sensor (HC-SR04) is used to detect water level in the overhead water tank (refer Fig. 11). Height of the tank used in the prototype is 20cm. If distance measured by the sensor is 0cm, the app will show the tank as full; and if distance is 20cm, it will show the tank as empty.

As shown in the circuit, pins D0 and D1 of NodeMCU are connected to pins Trig and Echo of HC-SR04, respectively. Water level is monitored through HC-SR04 sensor, and water motor pump can be controlled through RL1, which is connected to pin D6 of NodeMCU.

DHT11 sensor is connected to pin D2 of NodeMCU. Garden motor pump is controlled through RL4, which is connected to pin D3 of NodeMCU.

When power is switched on by connecting 12V battery, NodeMCU gets connected to Wi-Fi network. Open Blynk app on the mobile to start controlling the electrical appliances. But before that you need the required software in NodeMCU.

Arduino IDE is needed to write Arduino code and upload the programming logic onto NodeMCU board. Blynk app is required on the Android smartphone. Note that, authorisation code generated from Blynk app will be required in Arduino code. For that, create project in Blynk app.

Creating a project on Blynk IoT platform

Steps to create the project in Blynk are given below.

Install Blynk app on Android smartphone from Play Store. Register



Fig. 6: Set Tab2 for soil moisture level

yourself on the app. You will get an authentication token on your registered email ID submitted during the registration process. Note it down as you will need it for Arduino code later.

After registration, login to Blynk app and create your project by selecting New Project (Fig. 2). Name the project with a suitable title and select the module as NodeMCU (Fig. 3).

Now, open widget and select Buttons (Fig. 4). You will now see another screen where you can add label value to display temperature, humidity, super chart for graphical representation, etc.

Add two control tabs: Tab1 and Tab2 (Fig. 5). Under Tab1, add Light1 and Light2 for controlling Load3 and Load2, respectively. Under Tab2, add two levels, V7 and V8, to display soil resistance and water level, respectively. You can also add eventor and notification to get notifications on a particular event.

Now, link all controls in Blynk app and pins of NodeMCU, as given in Table above.

The author's prototype is shown in Fig. 8.



Fig. 7: Eventor settings

Various Functions of Blynk App and NodeMCU Pins

Blynk App	Function
Tab1:Load3	D4 pin (NodeMCU)
Tab1:Load2	D5 pin (NodeMCU)
Motor	D3 pin (NodeMCU)
Tab2:V7	Virtual moisture level
Tab2:V8	Virtual water level

Next, open `smart_home.ino` code from Arduino IDE. Configure NodeMCU with Wi-Fi credentials and paste the authorisation code generated from Blynk app. Compile the code and upload it onto NodeMCU board.

Construction and testing

You can assemble the circuit on a veroboard or on the designed PCB given here. An actual-size PCB layout for the IoT-based device controller for smart home is shown in Fig. 9 and its components layout in Fig. 10.

After assembling the circuit, enclose it in a suitable plastic box. Power on the circuit. Open the project from Blynk app to check whether all devices are connected to the Internet. If they are connected, red mark on the app will disappear.

You can start controlling the devices by touching the corresponding buttons on the app. Garden motor